ApexPlanet Data Analytics Internship

Task 4: Data Storytelling & Statistical Validation

Customer Segmentation & Business Insights

Prepared By

Alok Kumar

B.Tech – Computer Science & Engineering

Indian Institute of Information Technology (IIIT) Kottayam

ApexPlanet
Software Pvt.Ltd

Agenda

01. Project Overview

02. Dataset Overview

03. Data Cleaning & Exploratory Analysis

04. Customer Segmentation

05. Business Story & Key Insights

06. Hypothesis Testing

07. Business Recommendations

08. Conclusion

Project Overview

Project Objective

The internship aims to transform raw data into meaningful business insights through:

- Data Cleaning & Preparation
- Exploratory Data Analysis (EDA)
- SQL Analysis
- Customer Segmentation
- Interactive Dashboard Development
- Statistical Validation

- Portfolio Development

Project Scope

The internship consisted of **five major tasks:**

- **Task 1** – Data Cleaning & Preparation
- **Task 2** – Exploratory Data Analysis (EDA), SQL Analysis & Tableau Dashboard
- **Task 3** – Customer Segmentation & Interactive Dashboard
- **Task 4** – Data Storytelling & Statistical Validation
- **Task 5** – Capstone Integration & Portfolio Finalization

Tools & Technologies

- Python
- Pandas
- Matplotlib
- MySQL
- Tableau Public
- SciPy
- Git & GitHub
- Visual Studio Code

Expected Outcome

- Develop practical data analytics skills.
- Generate actionable business insights for data-driven decision-making.
- Build a professional portfolio showcasing all internship projects.

Dataset Overview

Dataset Name

ApexPlanet Retail Sales Dataset

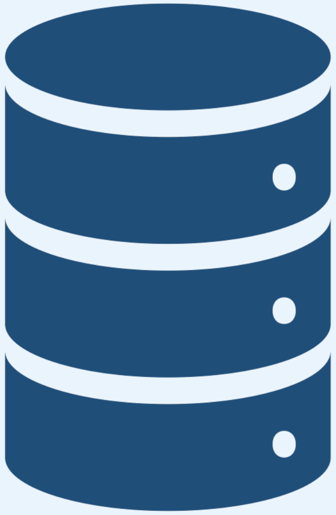
Records

1,000 Sales Records

7 Key Attributes

Dataset Includes

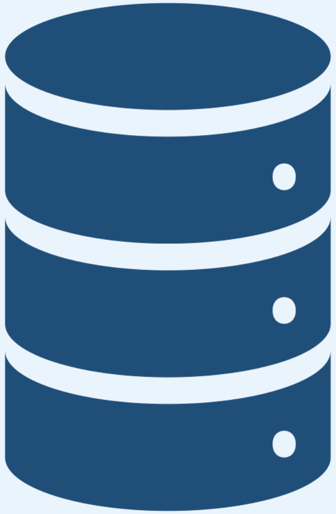
- Customer Information
- Product Details



- Sales Data
- Pricing Information
- Purchase History
- Customer City
- Product Category

Business Goal

- Analyze customer purchasing behaviour.
- Identify purchasing patterns.
- Discover sales trends.
- Support data-driven decision making.
- Identify growth opportunities.



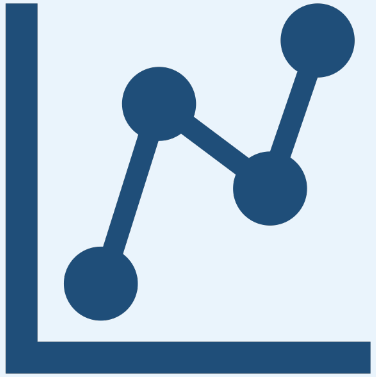
Data Cleaning & Exploratory Analysis

Task 1 – Data Cleaning

- Missing values handled
- Duplicate records removed
- Data types standardized
- Feature engineering completed
- Dataset prepared for analysis

Task 2 – Exploratory Data Analysis

- Descriptive Statistics
- Correlation Analysis
- Distribution Analysis
- Business SQL Queries
- Interactive Tableau Dashboard



Customer Segmentation

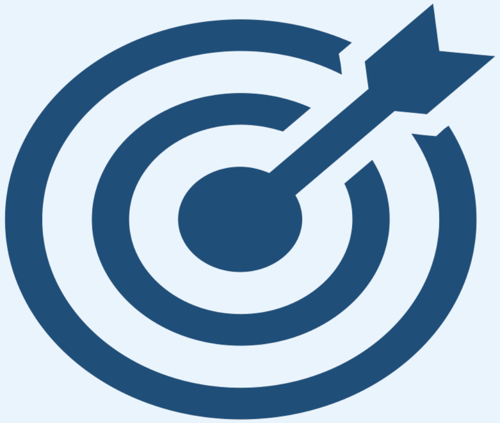
Customer Segments Identified

- Champions
- Big Spenders (At Risk)
- Big Spenders (New)
- Loyal Regulars
- Recent One-Time Buyers
- Standard One-Time Buyers

- Lapsed Customers

Key Finding

- Customer spending varies across segments.
- Champions contribute the highest revenue.
- Lapsed customers require re-engagement.
- Customer segmentation enables targeted marketing.
- Helps improve customer retention and revenue growth.



Business Story & Key Insights

Business Insights

- Total revenue exceeded **₹139 million**.
- Electronics generated the highest revenue.
- Customer purchasing behaviour varied across different segments.
- **Big Spenders (At Risk)** represent a high-value retention opportunity.
- Interactive Tableau dashboards enabled data-driven business analysis.

Business Impact

- Improved visibility into customer purchasing behaviour.
- Enabled data-driven business decisions.
- Identified high-value customer segments.
- Supported targeted marketing strategies.
- Helped uncover revenue growth opportunities.

Hypothesis Testing

Business Hypothesis

Null Hypothesis (H_0)

There is no statistically significant difference in the average spending between **Loyal Regulars** and the **combined Big Spenders group (Big Spenders (At Risk) + Big Spenders (New)).**

Alternative Hypothesis (H_1)

There is a statistically significant difference in the average spending between **Loyal Regulars** and the **combined Big Spenders group (Big Spenders (At Risk) + Big Spenders (New)).**

Statistical Method

- **Test Used:** Welch's Independent Samples t-test
- **Confidence Level:** 95%
- **Significance Level (α):** 0.05
- **Groups Compared:** Loyal Regulars vs Combined Big Spenders (At Risk + New)
- **Objective:** Determine whether the average spending differs significantly between the two customer groups.

Results & Recommendations

Statistical Result

- p-value < 0.05
- Reject the Null Hypothesis (H_0)
- The difference in average spending is **statistically significant.**

Business Recommendations

- Launch personalized win-back campaigns for **Big Spenders (At Risk)**.
- Encourage **Big Spenders (New)** to become repeat customers through loyalty programs and personalized offers.

- Improve customer retention strategies.
- Continue monitoring KPIs through interactive dashboards.
- Use customer segmentation to optimize targeted marketing campaigns.

Conclusion

Conclusion

- Successfully transformed raw retail sales data into actionable business insights.
- Applied Python, SQL, Tableau, and statistical techniques to solve business problems.
- Identified customer segments and purchasing patterns.
- Built an interactive dashboard for data-driven decision-making.
- Validated business insights through hypothesis testing.

Thank You